Practical:

SUBJECT NAME: Machine Learning Lab **Credit: 2**

SUBJECT CODE: BCAC791B

After completing this lab, students will be able to implement basic machine learning workflows using suitable programming tools and libraries. They will prepare datasets through cleaning, preprocessing, and feature selection for model building. Students will apply regression, classification, clustering, and decision tree algorithms on real datasets and observe their behaviour. They will evaluate model performance using appropriate metrics and understand issues such as overfitting and underfitting. Learners will gain hands-on experience in training, testing, and validating models. Finally, students will interpret results and present meaningful insights through outputs, visualisations, and brief reports, demonstrating practical skills required for real-world machine learning tasks.

TOPICS TO BE COVERED

Module	Module Title	Practical Coverage in Python
1	Python Environment and Data Handling	Working in Jupyter/IDE; using NumPy and Pandas; importing CSV datasets; basic data operations
2	Data Preprocessing and EDA	Handling missing values; data cleaning; feature selection; train–test split; visualisation using Matplotlib
3	Regression Techniques	Implementing simple linear regression; prediction; calculating MAE, MSE, RMSE
4	Classification Techniques	Logistic Regression, KNN, Naïve Bayes; confusion matrix; accuracy, precision, recall
5	Decision Tree and Clustering	Building a decision tree model; implementing K-Means clustering; visualisation and interpretation
6	Model Evaluation and Mini Project	Cross-validation (basic idea); overfitting and underfitting; applying full ML workflow on a small dataset

SUGGESTED READING:

1. **"**Machine Learning using Python" by Manaranjan Pradhan & U. Dinesh Kumar – Wiley India *(2019; also 2nd edition 2025)*
2. "Machine Learning with Python" by Abhishek Vijayvargia – BPB Publications *(2018)*
3. "Applied Machine Learning Solutions with Python" by Siddhanta Bhatta – BPB Publications *(2022)*
4. "Introduction to Machine Learning with Python: A Guide for Data Scientists" by Andreas C. Müller & Sarah Guido (Greyscale Indian Edition) – Shroff Publishers *(2016)*
5. "Data Science and Machine Learning Using Python" by Reema Thareja – McGraw Hill India